Exoplanet Habitability Prediction System

Backend Documentation

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Module: Backend Development

Technology: Python, Flask, Machine Learning

Project Overview

This project is a machine learning based system that predicts the habitability of exoplanets.

The backend is built using Flask which exposes APIs for prediction and ranking.

The system analyzes astronomical parameters such as orbital period, planetary radius, planet mass and stellar properties to determine whether a planet may support life.

Tech Stack

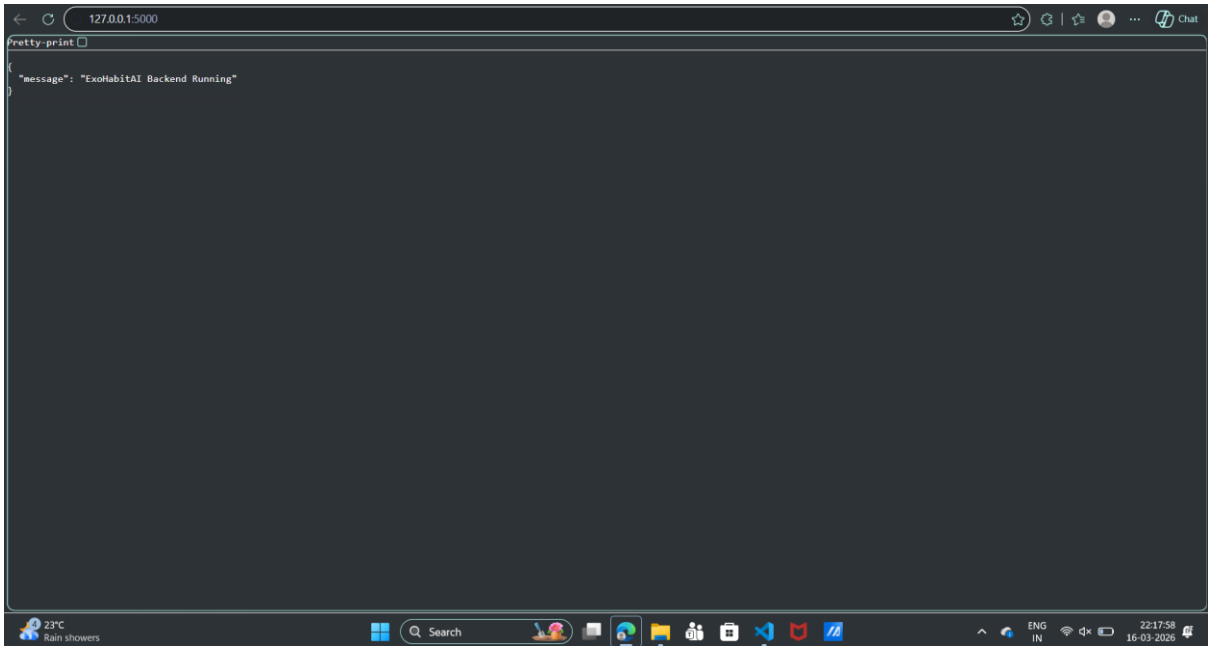
Programming Language: Python

Framework: Flask

Libraries:

- Pandas
- Scikit-learn
- NumPy

Flask Server



Predict API

Endpoint: /predict

Method: POST

Input:

{

"features":[365,1.2,5.1,1.0,280,5700,1.0,1.0,1.0]

}

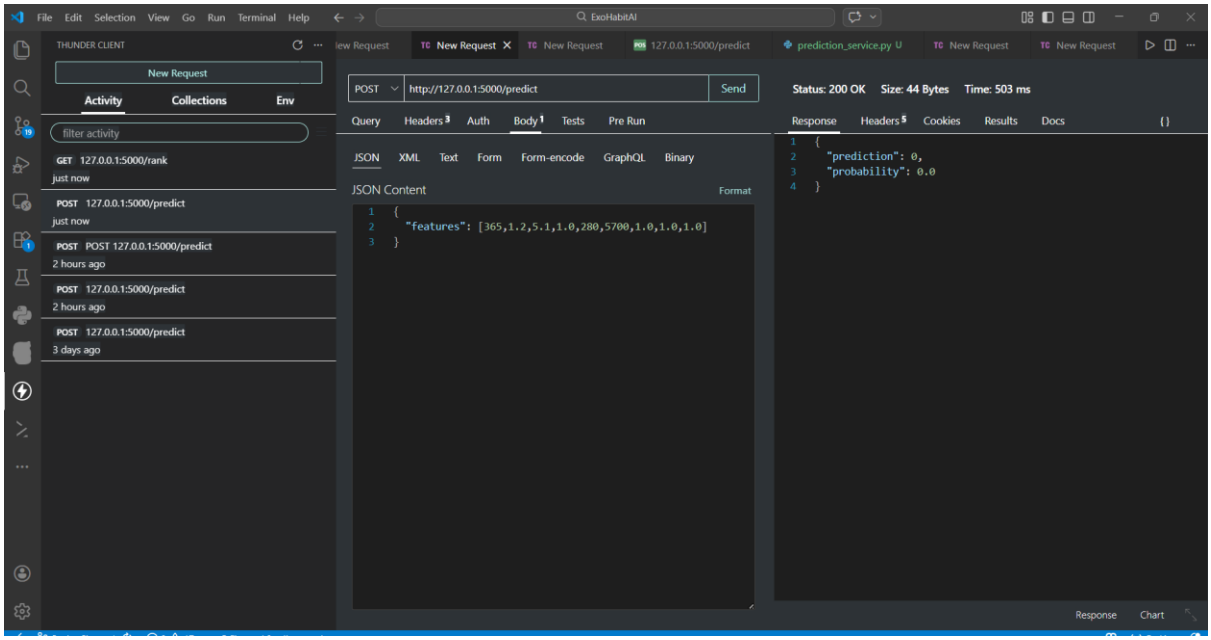
Output:

{

"prediction":0,

"confidence":0.0

}



Rank API

Endpoint: /rank

Method: GET

Output:

{

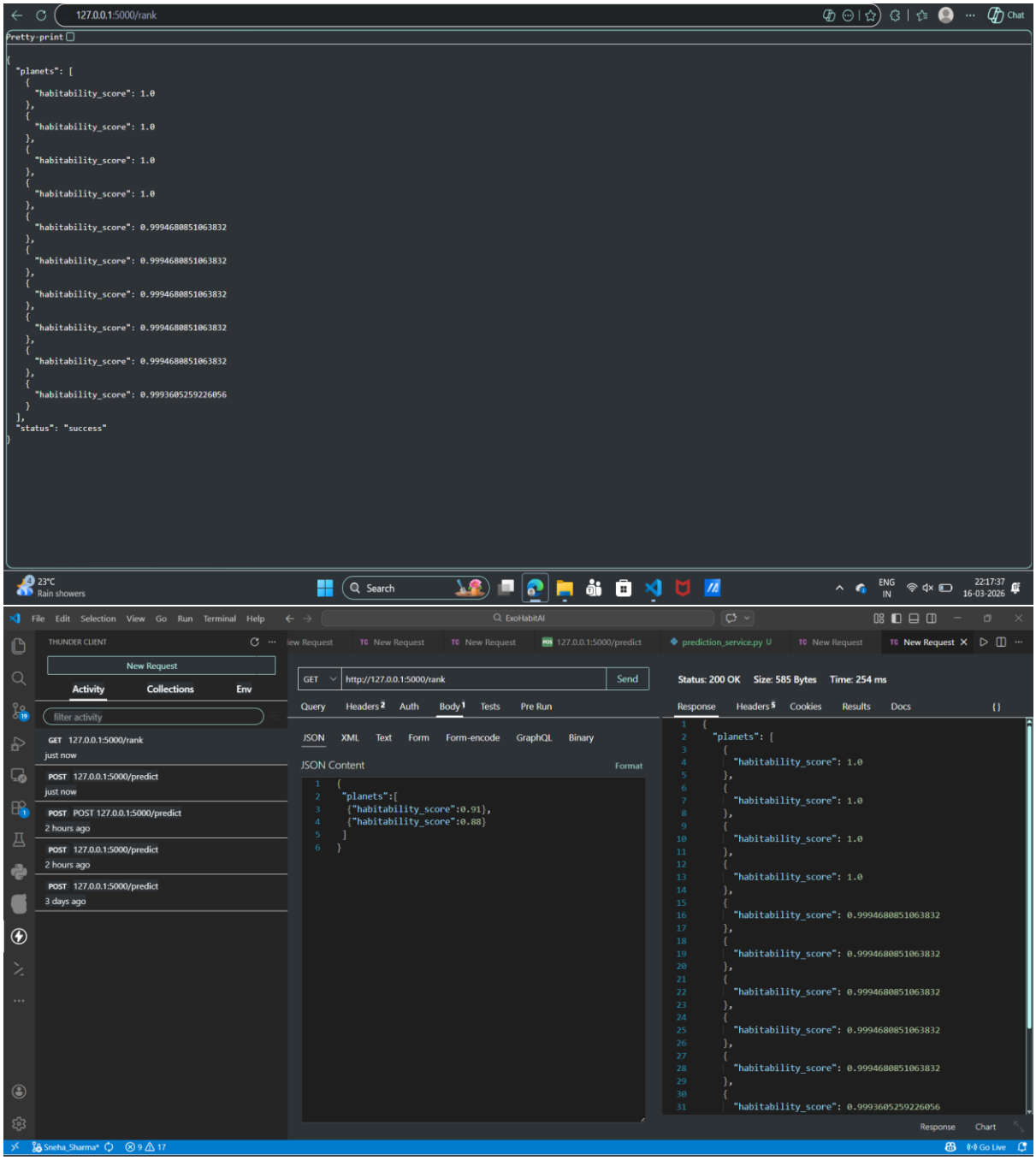
"planets":[

{"habitability_score":0.91},

{"habitability_score":0.88}

]

}



Input Validation

The backend validates incoming JSON requests.

If required parameters such as features are missing,

the API returns an appropriate error message.

i.e - {

"error":"features missing"

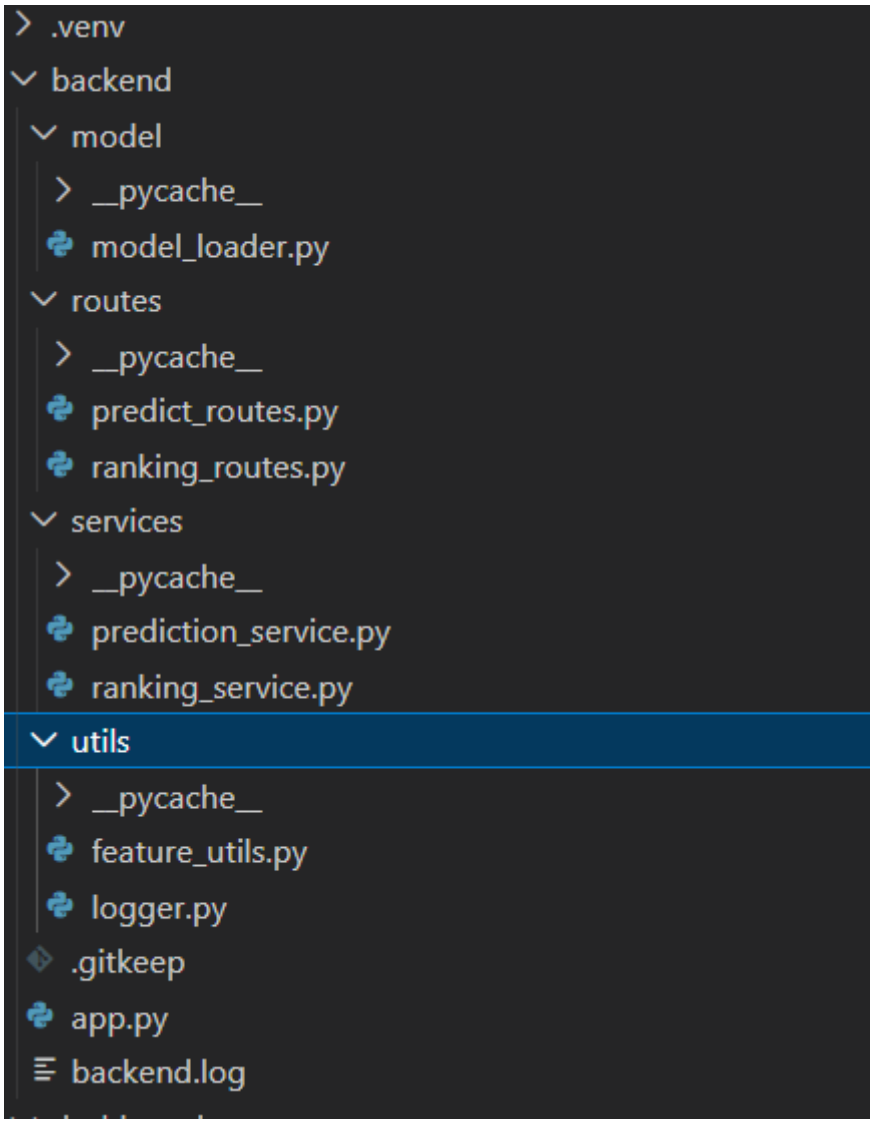
}

Testing

The APIs were tested using:

- Browser
- Thunder Client
- JSON requests

Project Structure : Backend



Result

The backend successfully integrates a machine learning model to predict the habitability of exoplanets.

Two APIs were implemented:

1. Prediction API
2. Planet Ranking API